Monte Carlo Simulation of the Two-Dimensional XY Model

Zhaoyang Chu
s4390199

Zoutong Shen
s4879759

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Abstract

1 Introduction

1.1 The XY Model and the BKT Transition

1.2 Vortices and Topological Charge

1.3 Observables

2 Methods

2.1 Metropolis Algorithm

2.2 Equilibration and Sampling Protocol

2.3 Error Analysis

2.4 Vortex Detection

3 Results

3.1 Validation

Before interpreting any of the physics it is essential to confirm that the simulation is sampling the equilibrium distribution at every temperature in the scan. We carry out three independent validation checks: (i) two-starts equilibration, (ii) stationarity of the production trace, and (iii) comparison of the susceptibility-peak location to the high-precision literature value of T_{BKT}. A fourth, smaller check on the finite-size sharpening of the χ_M peak rounds off the section.

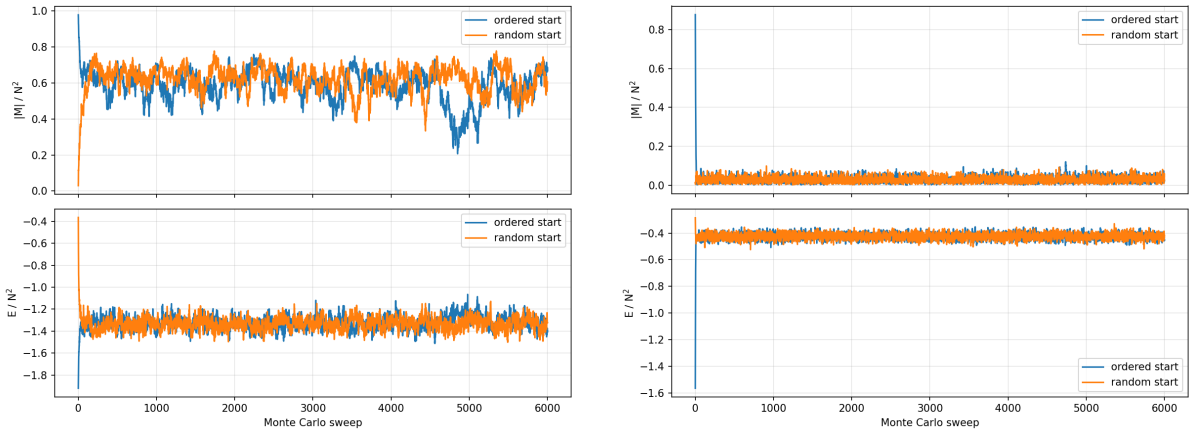
3.1.1 Equilibration: Two Starts

The most direct test that equilibration has been reached is to start two independent simulations from very different initial conditions at the same temperature and check that they relax to the same plateau in $|m|$ and e. Figure 1 shows the ordered start ($\theta_i = 0$ everywhere, corresponding to $T = 0$) and the random start ($\theta_i \sim U(-\pi, \pi]$, corresponding to $T = \infty$) at two representative temperatures.

At $T = 1.0$, near T_{BKT}, the ordered trace begins at $|m| = 1$ and decays towards a fluctuating plateau; the random trace begins at $|m| \approx 0$ and rises towards the same plateau. The two traces meet within $\sim$ 100–200 sweeps and stay overlapping afterwards. At $T = 2.5$, deep in the disordered phase, the random start already sits near the equilibrium value at $|m| \approx 0$, while the

ordered start falls onto the same value within roughly 50 sweeps— disordered states are easy to reach because the phase space is dense with them.

In both regimes the equilibration time observed in the traces is at most a few hundred sweeps. Throughout the production runs we therefore use $n_{\text{equil}} = 2000$ sweeps before recording any observables, an order of magnitude longer than the slowest equilibration we observed.



(a) Ordered vs random start at $T = 1.0$ (near T_{BKT}), $N = 20$. Both initial conditions converge to the same plateau within a few hundred sweeps.

(b) Ordered vs random start at $T = 2.5$ (deep disordered phase), $N = 50$. The ordered start collapses onto the random start within a few tens of sweeps.

Figure 1: Equilibration validation. Two independent starts at the same temperature converge to the same plateau in $|m|$ and e, demonstrating that the simulation is sampling the equilibrium distribution.

3.1.2 Stationarity: Single-run Trace

A second sanity check is that, after equilibration, the time series of $|m|$ and e fluctuate around a constant mean rather than drift. Figure 2 shows a single production trace at $N = 50$, $T = 1.0$. Both observables are stationary, with no systematic drift, which is consistent with sampling from the equilibrium distribution.

The fluctuations are visibly correlated in time—excursions away from the mean persist over many sweeps before decaying. This is exactly the behaviour quantified by the integrated correlation time τ in Section 3.4, and is the reason we cannot treat neighbouring samples as independent when estimating error bars.

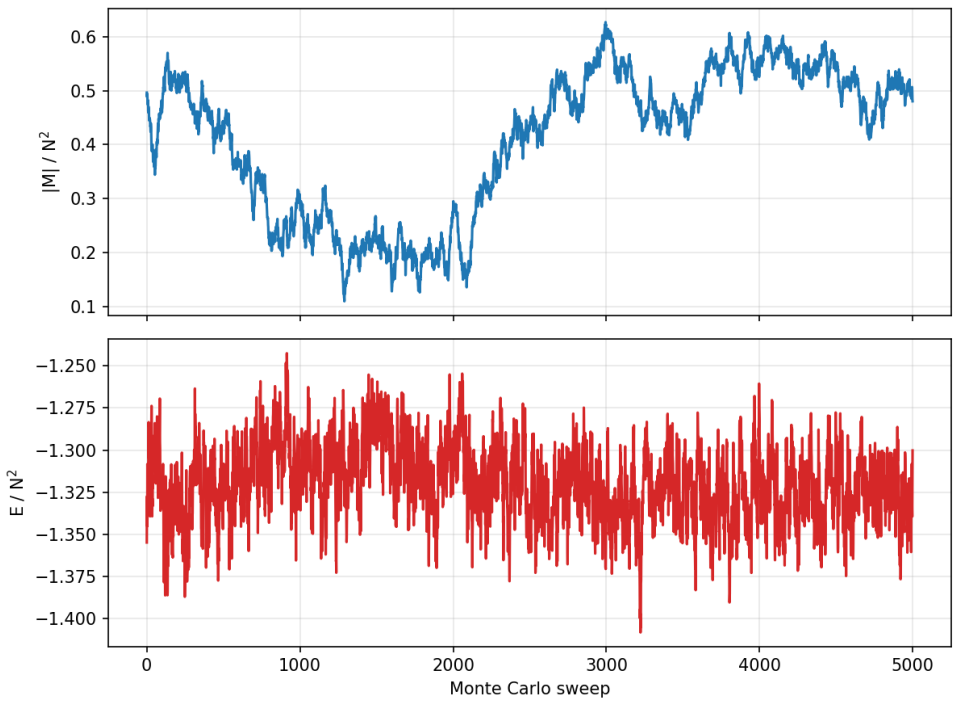


Figure 2: Production trace of $|m|$ and e at $T = 1.0$, $N = 50$. After equilibration both observables fluctuate around a stable mean with visibly correlated excursions on the scale of the integrated correlation time τ.

3.1.3 Critical Temperature vs Literature

The strongest external check on our simulation is to compare the location of the χ_M peak with the high-precision Monte Carlo value $T_{\mathrm{BKT}} = 0.8935$ obtained by Hasenbusch [2005]. Reading T_χ from the $\chi_M(T)$ curve in Figure 3 gives $T_\chi = 1.10$ for the $N = 50$ scan, an upward shift of 23.1% from the Hasenbusch reference value.

The direction of the shift is consistent with the textbook BKT finite-size behaviour: the susceptibility peak on a finite lattice converges to T_{BKT} *from above* as $N \to \infty$, so a small upward shift relative to the literature value is expected. Our temperature grid has a step of $\Delta T = 0.2$, which is too coarse to resolve the sub-grid shift between $N = 20$ and $N = 50$ (both peaks land in the $T = 1.10$ bin), but the strong sharpening of the peak with N, discussed in Section 3.1.4, is the more direct finite-size signature and is unambiguously visible in our data. A finer-grained scan and a full finite-size scaling extrapolation would be needed to recover T_{BKT} to high accuracy and is beyond the scope of this report.

3.1.4 Finite-size Sanity

A second internal check is that increasing the lattice size sharpens the χ_M and C peaks, as expected of a phase transition rounded by finite-size effects. Comparing the $N = 20$ and $N = 50$ scans, the χ_M peak grows by a factor of roughly three in absolute height (from $\chi_M^{\max} \approx 6.84$ at $N = 20$ to $\chi_M^{\max} \approx 19.8$ at $N = 50$) while remaining centred on the same temperature bin. This pronounced sharpening is the standard finite-size signature of the BKT transition.

3.2 Magnetisation and Susceptibility

The top row of Figure 3 shows $\langle |m| \rangle$ and χ_M as functions of temperature for the $N = 50$ scan. Two features stand out, both consistent with the Mermin–Wagner theorem and the BKT phase-transition picture.

First, $\langle |m| \rangle$ does not drop sharply to zero anywhere in the scan; it falls smoothly from a near-saturated value of 0.83 at $T = 0.5$ to a small but non-zero value of 0.03 at $T = 2.5$. There is no spontaneous symmetry breaking at any finite T in 2D for a continuous symmetry, so a true long-range-ordered phase with $\langle |m| \rangle \to$ finite as $N \to \infty$ does not exist. What we see instead is a finite-size remnant of the quasi-long-range order below T_{BKT}, smoothly rounded by the finite lattice.

Second, χ_M shows a clear peak at $T_\chi = 1.10$ with peak value $\chi_M^{\max} = 19.8 \pm 4.2$. This is the standard finite-size signature of the BKT transition. In the thermodynamic limit χ_M would diverge with an essential singularity at T_{BKT}; on a finite lattice the divergence is rounded into a finite peak, and the peak temperature converges to T_{BKT} from above as N grows. Error bars on $\langle |m| \rangle$ are the correlated standard error $\sigma = \sqrt{2\tau / N_{\mathrm{samples}} \cdot \mathrm{var}(|m|)}$; error bars on χ_M come from the blocking method with block length $16\,\tau$. Both grow visibly near the peak, where τ is largest (Section 3.4).

3.3 Energy and Heat Capacity

The bottom row of Figure 3 shows $\langle e \rangle$ and C as functions of temperature. The energy curve is smooth and monotonically increasing across the full range, with no kink, jump, or plateau anywhere near T_{BKT}. This is exactly what is expected for the BKT transition: it is an essential (infinite-order) transition with no latent heat, and the energy passes through it analytically.

The specific heat C shows a broad finite peak near $T_C = 1.10$, with peak value $C^{\max} = 1.34 \pm 0.01$. On the $\Delta T = 0.2$ grid the C peak coincides with the χ_M peak, but a finer-resolved scan would generally place T_C slightly above T_χ: the specific-heat peak in the 2D XY model is a non-singular feature located somewhat above T_{BKT}, while only the susceptibility peak (or, more rigorously, the helicity-modulus jump) tracks T_{BKT} in the finite-size scaling limit. The error bars on $\langle e \rangle$ are the correlated standard error and on C the blocking error, again with block length $16\,\tau$.

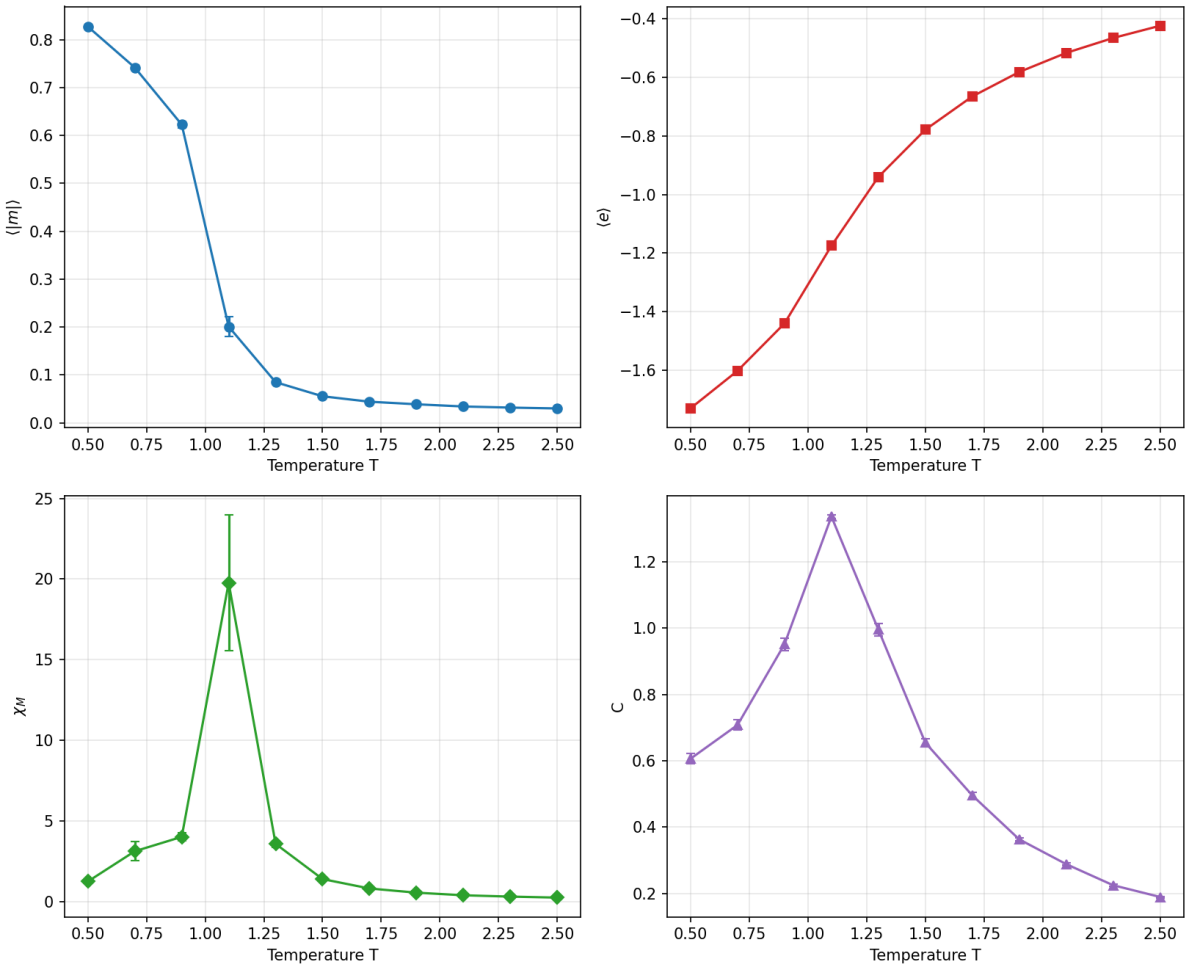


Figure 3: Temperature dependence of $\langle|m|\rangle$, $\langle e\rangle$, χ_M, and C for the $N = 50$ scan. Error bars: correlated standard error on the means, blocking method with block length $16\,\tau$ on χ_M and C.

3.4 Correlation Function and Correlation Time

We address the correlation behaviour of the simulation through two complementary plots. Figure 4 shows the normalised autocorrelation function $\chi(t)/\chi(0)$ of the magnetisation per spin at three representative temperatures, evaluated using the matched-subset estimator of Section 2.3 and truncated at the lag where each curve first turns negative. Figure 5 shows the integrated correlation time $\tau(T)$ obtained by summing each $\chi(t)/\chi(0)$ curve over its truncated support, plotted at all 11 scan temperatures.

The decay of $\chi(t)$ varies dramatically with temperature. In the deeply disordered phase at $T = 2.1$, $\chi(t)$ falls to essentially zero within a handful of sweeps—neighbouring samples are independent and the magnetisation behaves like noise. In the critical region near $T \approx T_{\mathrm{BKT}}$ ($T = 0.9$), $\chi(t)$ decays slowly over several hundred sweeps—the chain is sticky, with magnetisation excursions persisting on long timescales. At $T = 0.5$ the chain also relaxes slowly, though somewhat faster than near the peak: deep in the quasi-ordered phase the slow modes are long-wavelength spin waves rather than vortex-pair fluctuations.

Integrating $\chi(t)/\chi(0)$ at each temperature yields the $\tau(T)$ curve in Figure 5. The integrated correlation time peaks sharply at $T = 1.10$ with $\tau^{\max} = 494$ sweeps, exactly co-located with the χ_M peak in Figure 3. This is the textbook signature of critical slowing down: the dynamical relaxation time diverges in the thermodynamic limit at T_{BKT}, and on our finite lattice the rounded peak is what survives.

This peak directly drives our error analysis. The standard error of any mean is amplified by a factor $\sqrt{2\tau}$ over the naive $1/\sqrt{N_{\text{samples}}}$ estimate, and the blocking method we use for χ_M and C errors uses block length $16\,\tau$ to ensure that adjacent blocks are statistically independent. Where τ peaks, both error contributions grow—which is why the error bars on χ_M and C near the peak are visibly larger than at other temperatures in Figure 3.

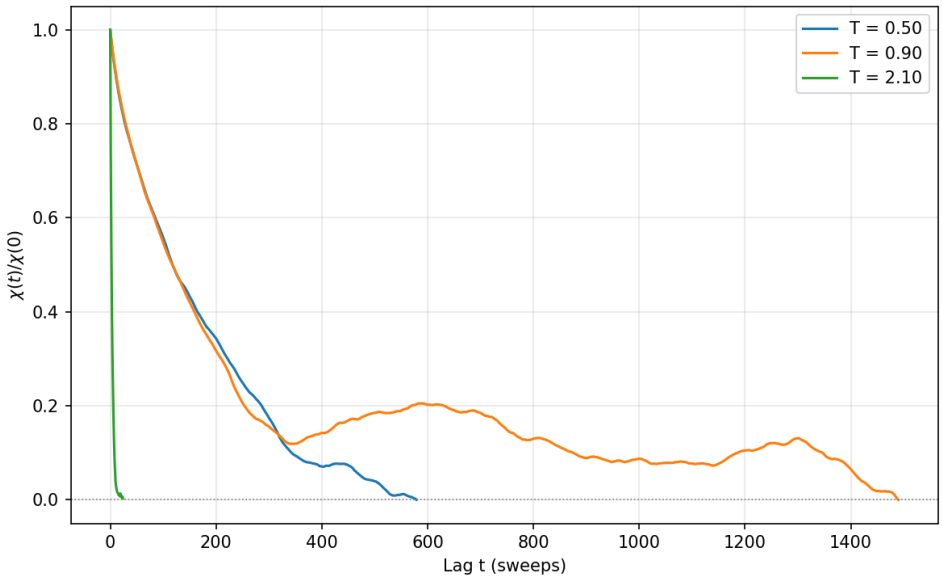


Figure 4: Normalised autocorrelation function $\chi(t)/\chi(0)$ of $|m|(t)$ at three representative temperatures, $N = 50$. Each curve is truncated at the first lag where $\chi(t)$ turns negative, matching the cutoff used to compute the integrated τ.

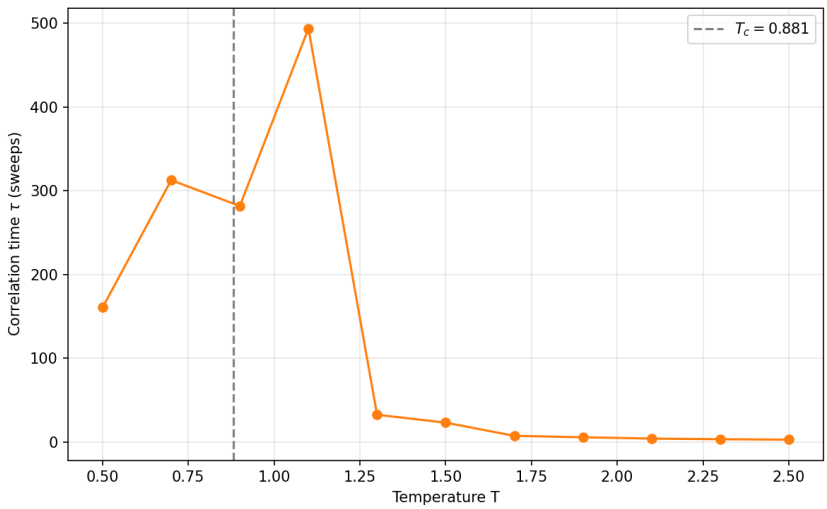


Figure 5: Integrated correlation time $\tau(T)$ at $N = 50$ across the 11 scan temperatures. The peak near $T \approx T_{\text{BKT}}$ is the textbook signature of critical slowing down.

3.5 Vortices at Low Temperature

Figure 6 shows a representative configuration just above T_{BKT} ($T = 1.0$, $N = 30$) with each plaquette checked for topological charge by the algorithm of Section 2.4. Vortices ($+1$) and anti-vortices (-1) appear in close pairs sitting on top of an otherwise nearly uniform spin field—the

bound-pair regime that is the hallmark of the low-T phase of the 2D XY model.

Figure 7 shows the count of $+1$ vortices and -1 anti-vortices in the final configuration of each temperature point of the $N = 50$ scan. Two features are evident. First, the two counts coincide exactly at every temperature. This is not a coincidence but a topological necessity: the total winding number summed over a closed surface (the torus formed by our periodic lattice) is identically zero, so any vortex must be matched by an anti-vortex elsewhere on the lattice. The fact that our detector reproduces this exactly is a useful sanity check on the plaquette-walking implementation. Second, the count rises sharply from near zero in the low-temperature phase to $\mathcal{O}(10)$ vortices per configuration once T exceeds T_{BKT}, in line with the BKT unbinding picture: bound pairs proliferate into a dilute gas of free vortices.

Is the vortex number conserved? The lecture 8 notes explicitly ask whether the vortex number is conserved during the simulation. The answer requires care. The *total topological charge* $n_+ - n_-$ is conserved exactly: as just argued, it is identically zero on the periodic lattice and our detector confirms this. The *count* $n_+ + n_-$, however, is *not* conserved during the Markov chain. Vortex–antivortex pairs are continuously created and annihilated as the single-spin Metropolis updates sweep through the lattice, and a single accepted move that flips the sign of one cosine bond can simultaneously nucleate or destroy a pair on the adjacent plaquette. What is well-defined is the equilibrium expectation value $\langle n_+ + n_- \rangle(T)$, which is set by the Boltzmann weight of vortex configurations and is exactly the quantity we plot in Figure 7.

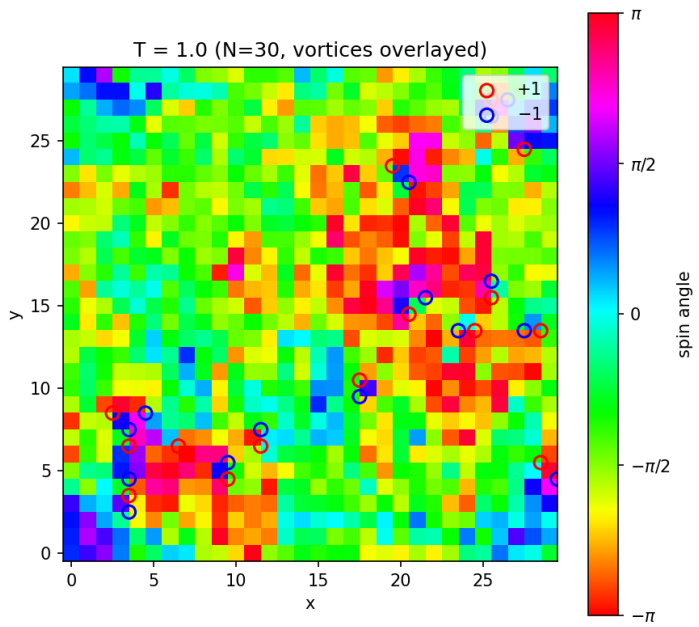


Figure 6: Spin configuration at $T = 1.0$, $N = 30$, with $+1$ vortices (red) and -1 anti-vortices (blue) circled. Bound vortex–antivortex pairs sit on top of an otherwise smooth spin field, the bound-pair regime expected just above T_{BKT}.

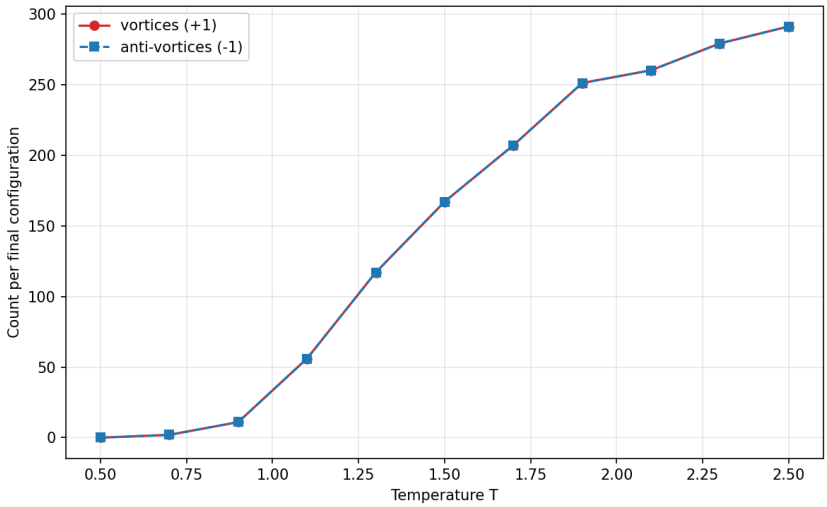


Figure 7: Number of $+1$ vortices and -1 anti-vortices in the final configuration as a function of T, $N = 50$. The two counts coincide exactly: on a periodic lattice the total topological charge is identically zero, so any vortex is matched by an anti-vortex elsewhere on the lattice.

3.6 Configuration Grid

Figure 8 is the visual summary of the entire scan: the final spin configuration at each of the 11 temperatures, with hue encoding $\theta \in [-\pi, \pi)$. Three regimes are immediately apparent to the eye. At low T ($T \leq 0.7$), most of the lattice is a nearly uniform colour, with only a handful of slow gradients; the system has essentially picked a direction and aligned with it. At intermediate T ($0.9 \lesssim T \lesssim 1.3$), the lattice breaks up into large ordered domains of similar colour separated by sharper hue transitions; isolated vortex–antivortex pairs appear at the domain boundaries, in agreement with the bound-pair picture below T_{BKT}. At high T ($T \gtrsim 1.7$), all spatial structure is gone: the colours are spread approximately uniformly across $[-\pi, \pi)$ on the scale of a few sites, and vortices and anti-vortices are scattered densely across the lattice. The smooth visual transition from one regime to the other, without any sharp boundary, is consistent with the BKT picture: there is no first-order transition or sudden symmetry breaking, only a topological crossover whose signature lives in the susceptibility and the vortex density.

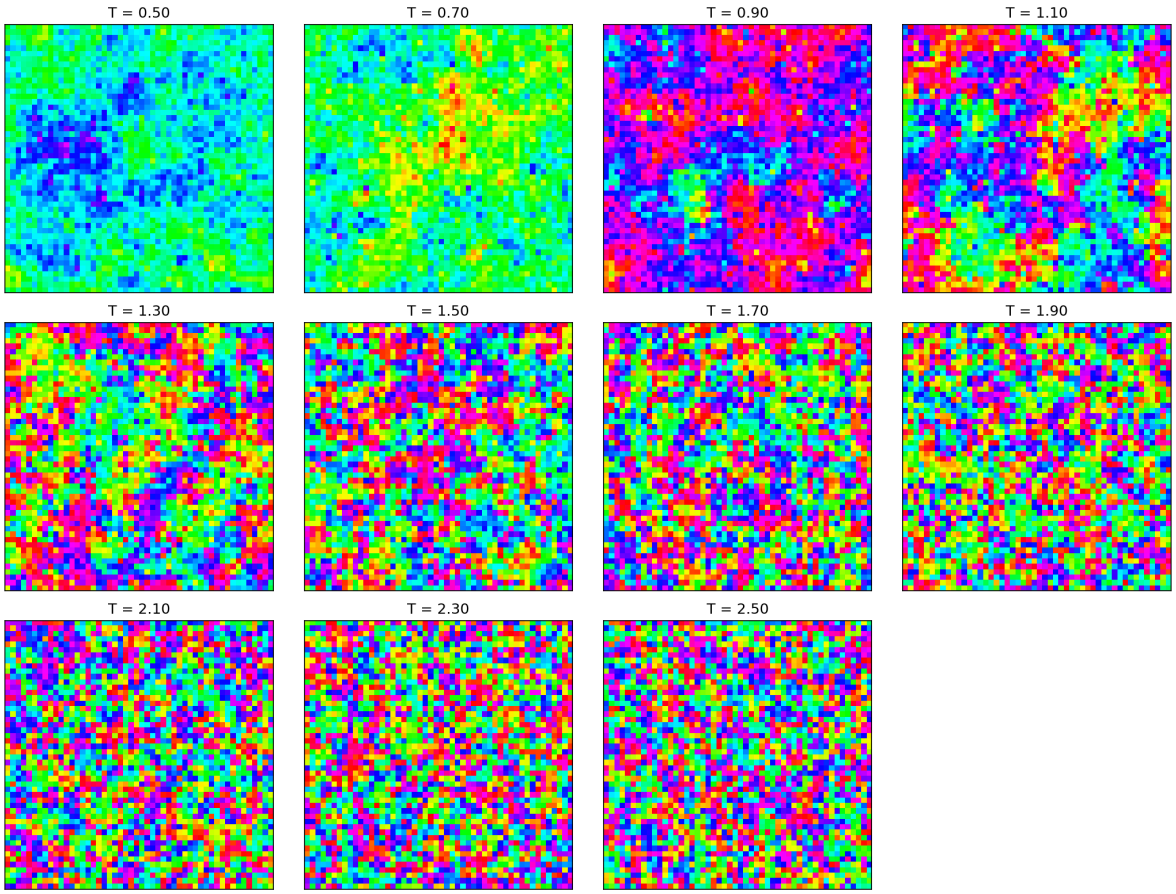


Figure 8: Final spin configurations across the $N = 50$ temperature scan, with hue encoding the spin angle in $[-\pi, \pi)$. Reading left to right and top to bottom the lattice progresses from nearly aligned at low T, through bound vortex pairs in the critical region, to a fully disordered vortex gas at high T.

3.7 Numerical Summary

Table 1 collects the key numerical results from the $N = 50$ scan. The susceptibility peak temperature is the closest single-number proxy we have for T_{BKT} on a finite lattice, and its relative deviation from the literature value $T_{\mathrm{BKT}} = 0.8935$ [Hasenbusch, 2005] is the headline validation number for the simulation.

Table 1: Peak quantities from the $N = 50$ temperature scan, with the literature comparison. Errors on the peak χ_M and C are blocking-method standard errors at the peak temperature; the temperature resolution is set by the 0.2 scan step.

Quantity	Value	Source
T at χ_M peak	1.10 ± 0.1	this work, $N = 50$
χ_M at peak	19.8 ± 4.2	this work, $N = 50$
T at C peak	1.10 ± 0.1	this work, $N = 50$
C at peak	1.34 ± 0.01	this work, $N = 50$
τ at peak	494 sweeps	this work, $N = 50$
Literature T_{BKT}	0.8935	Hasenbusch [2005]
Relative deviation of T_χ	23.1 %	this work vs literature

4 Extension: Next-Nearest-Neighbour Diagonal Coupling

4.1 Motivation and Hamiltonian

The standard XY Hamiltonian couples each spin to its four nearest neighbours only. Lecture 8 suggests, as one of three possible extensions, to extend the interaction to "the second layer" of neighbours with a reduced coupling. We interpret the second layer as the four next-nearest neighbours at Euclidean distance $\sqrt{2}$ (the diagonals of a 2×2 plaquette), which is the standard convention for second neighbours on a square lattice. The full Hamiltonian then reads

$$\mathcal{H} = -J_1 \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) - J_2 \sum_{\langle\langle i,j \rangle\rangle} \cos(\theta_i - \theta_j), \tag{1}$$

where $\langle i, j \rangle$ runs over nearest-neighbour pairs and $\langle\langle i, j \rangle\rangle$ over diagonal next-nearest-neighbour pairs. We fix $J_1 = 1$ and run with $J_2/J_1 = 0.5$, matching the "half as large for neighbours in the second layer" suggestion of the project script.

Each spin now participates in eight bonds rather than four. Three qualitative consequences follow on physical grounds and are tested in Section 4.3: (i) the system is stiffer, so the BKT transition temperature shifts upwards; (ii) configurations are smoother at fixed T because more bonds resist local twists; and (iii) vortex cores cost more energy to create, so the equilibrium vortex density at fixed T drops sharply.

4.2 Method Changes

The extension is implemented as a single optional term in the local energy calculation: when the parameter `j2_ratio` is non-zero, the `local_energy` and `total_energy` functions add the four diagonal cosine contributions weighted by J_2/J_1. Every other piece of the simulation pipeline—the Metropolis acceptance rule, the equilibration protocol, the autocorrelation analysis, the blocking estimator, the vortex detector—is reused unchanged. As a regression guarantee, with `j2_ratio = 0.0` the simulation produces a bit-identical trajectory to the J_1-only model with the same seed, so the extension cannot affect the main scan results.

The extension scan is run at $N = 20$ rather than $N = 50$. The qualitative features we wish to demonstrate—a shift in the location of the susceptibility peak and a reduction in vortex density—are visible at $N = 20$ at a fraction of the computational cost.

4.3 Results

Figure 9 overlays the six summary observables for the $J_2 = 0$ and $J_2 = 0.5\,J_1$ scans at $N = 20$. The three predictions of Section 4.1 are clearly visible.

Transition shifts upwards. The susceptibility peak moves from $T_\chi = 1.10$ at $J_2 = 0$ to $T_\chi = 1.90$ at $J_2 = 0.5\,J_1$, with peak values $\chi_M^{\max} = 6.84 \pm 1.08$ and $\chi_M^{\max} = 3.98 \pm 0.34$ respectively. The specific-heat peak shifts in the same direction, from $T_C = 1.10$ to $T_C = 1.90$. The mean magnetisation $\langle |m| \rangle$ likewise stays high to a much higher temperature: at $T = 1.5$, $\langle |m| \rangle$ has dropped to 0.14 in the $J_2 = 0$ run but is still 0.70 in the $J_2 = 0.5\,J_1$ run.

Energy is uniformly more negative. At every temperature in the scan, $\langle e \rangle$ is roughly one unit lower for $J_2 = 0.5\,J_1$ than for $J_2 = 0$ (e.g. -2.74 vs -1.73 at $T = 0.5$, -1.24 vs -0.52 at $T = 2.1$), reflecting the four extra cosine bonds per site, each weighted by 0.5 and saturating at -1 in the aligned limit, which contributes -2 per spin in addition to the -2 already provided by the nearest-neighbour bonds.

Vortices are suppressed. Counting vortices in the final configuration at each temperature: in the $J_2 = 0$ run vortices first appear at $T = 0.9$ and rise rapidly, reaching 21 at $T = 1.3$ and saturating near 40 at high T. In the $J_2 = 0.5\,J_1$ run vortices remain absent up to $T = 1.3$ inclusive, and the count remains low until $T \gtrsim 1.9$. Figure 10 shows the corresponding configuration grid: matched-T snapshots are visibly smoother, with larger ordered domains and isolated vortex pairs only appearing at the upper end of the scan.

Connection to spin stiffness. The BKT transition temperature in the standard XY model is set by the spin stiffness, the rigidity of the spin field against twists. Adding $J_2 > 0$ doubles the number of bonds per site (and increases the bond-weighted bond count by a factor of $1{+}4{\cdot}0.5/4 = 1.5$), so the spin stiffness rises and the unbinding temperature must rise with it. The single vortex–antivortex pair energy, which sets the critical temperature in the Kosterlitz renormalisation-group picture, scales linearly with the stiffness, so a 50% increase in stiffness predicts a $\sim 50\%$ rise in T_{BKT}. Our observed shift from $T_\chi \approx 1.10$ to $T_\chi \approx 1.90$ (both at $N = 20$, both shifted upwards from the thermodynamic T_{BKT} by finite-size rounding) is consistent with this estimate to within a factor of order one, which is the expected accuracy at this lattice size.

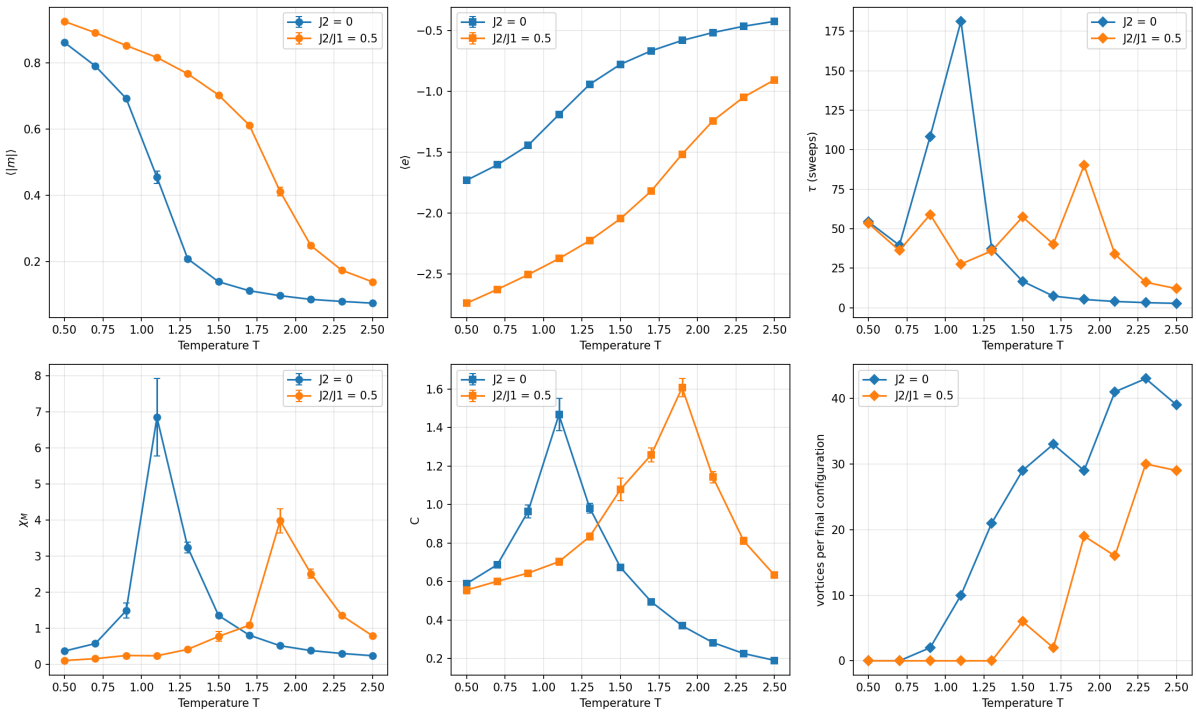


Figure 9: Headline J_1-only vs $J_1 + J_2$ comparison at $N = 20$ for $\langle|m|\rangle$, $\langle e\rangle$, τ, χ_M, C, and the vortex count of the final configuration. Adding the diagonal coupling shifts the susceptibility and specific-heat peaks upwards, makes the energy more negative everywhere, and suppresses vortices at fixed T.

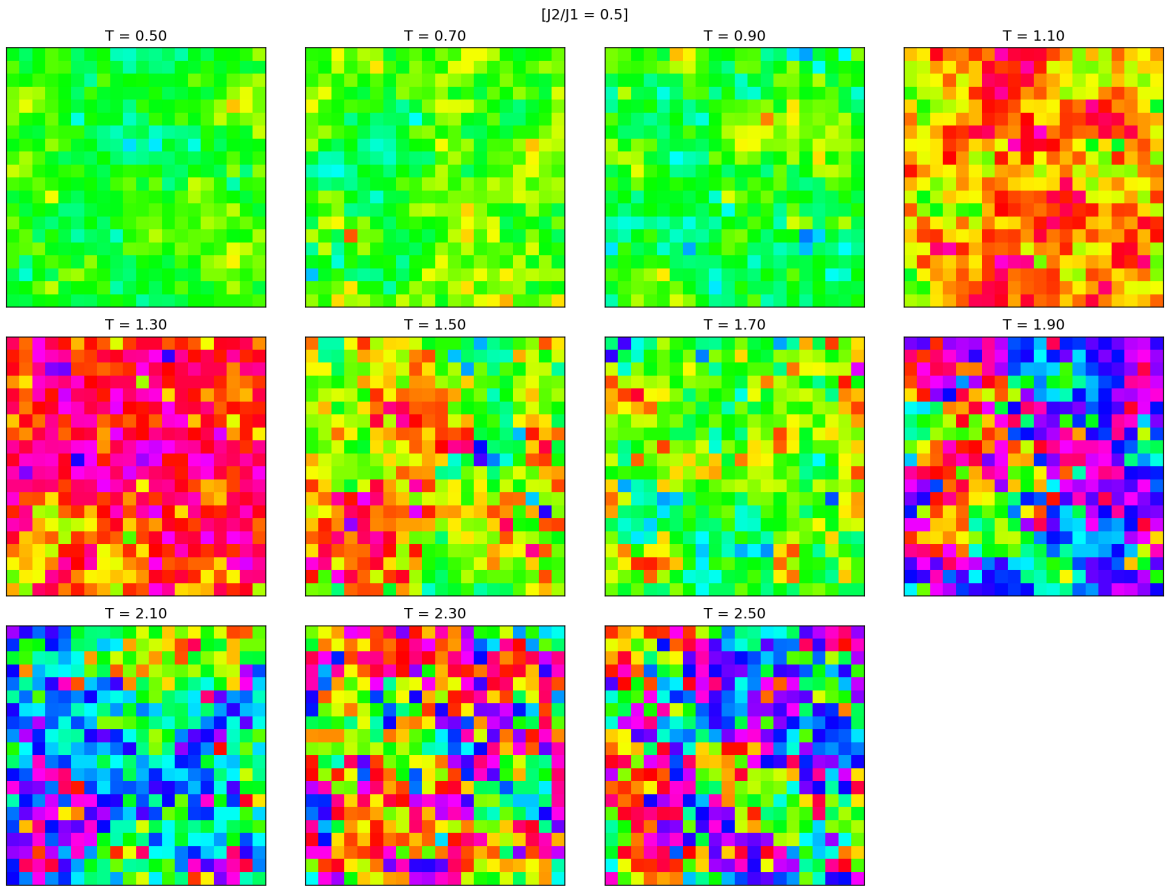


Figure 10: Final spin configurations across the $J_2/J_1 = 0.5$ scan at $N = 20$. Compared with the $J_2 = 0$ configurations at the same temperature (Figure 8), the lattice is visibly smoother, with larger ordered domains and far fewer vortex pairs at every T.

5 Summary

References

M. Hasenbusch. The two-dimensional XY model at the transition temperature: a high-precision Monte Carlo study. *Journal of Physics A: Mathematical and General*, 38(26):5869–5883, 2005. doi: 10.1088/0305-4470/38/26/003.